SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0603 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

D. Sindhu.

Signature of the student:

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labeled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

6. An autonomous vehicle system must make real-time driving decisions under uncertain environmental conditions such as weather, traffic, and pedestrian movement. Using Bayesian inference, Naïve Bayes classifiers, and probabilistic reasoning, design

a model to predict safe driving actions. Analyze how uncertainty and sensor noise are handled, and evaluate system reliability under dynamic conditions.

7. A healthcare organization aims to predict patient risk levels for chronic diseases using historical and lifestyle data. Apply probability learning, Bayes' Theorem, and hypothesis space search to design the model. Analyze the impact of prior probabilities, feature dependencies, and sample size. Evaluate the model's interpretability and its usefulness in preventive healthcare strategies.

8. A bank wants to detect fraudulent transactions using machine learning under uncertain and evolving patterns. Using Bayesian Belief Networks and EM algorithm concepts, design a detection system. Analyze hidden variables, incomplete data handling, and convergence issues. Evaluate how probability-based reasoning improves detection accuracy and reduces false positives in real-world scenarios.

9. An e-commerce platform aims to recommend products to users based on browsing and purchase history. Using Naïve Bayes and Bayesian learning concepts, design a recommendation model. Analyze how conditional probabilities influence recommendations and how uncertainty in user preferences is managed. Evaluate scalability and performance in handling large, dynamic datasets.

10. A research institute is building a system to predict climate changes using historical environmental data. Apply probabilistic models such as Bayesian inference and EM algorithm to handle incomplete and noisy datasets. Analyze convergence behavior, uncertainty modeling, and the influence of prior knowledge. Evaluate the system's effectiveness in long-term environmental forecasting.

11. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

12. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

13. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

14. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply **Naïve Bayes classifiers**, **EM algorithm**, and Bayesian

learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.

15. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using **Bayesian inference** and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Intelligent Disease Diagnosis System
Aim
To design a probabilistic model for disease diagnosis using Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks (BBN).
Approach

- Collect patient symptoms and medical history.
- Apply **Bayes' Theorem** to calculate the probability of diseases.
- Use a **Naïve Bayes classifier** for disease prediction.
- Use a **Bayesian Belief Network** to represent dependencies between symptoms and diseases.

Analysis

- Bayes' Theorem updates disease probability based on symptoms.
- Naïve Bayes assumes symptoms are independent.
- BBN models relationships between symptoms, diseases, and risk factors.
- Uncertainty is handled using conditional probabilities.

Advantages

- Fast and accurate prediction.
- Handles uncertain and incomplete medical data.
- Easy to update with new patient information.

Limitations

- Naïve Bayes assumes feature independence.
- Requires accurate probability values.
- Complex BBNs require more computation.

Result
The probabilistic model effectively predicts diseases while handling uncertainty, making it useful for clinical decision support.

2. Email Spam Detection System
Aim
To classify emails as spam or non-spam using Naïve Bayes and Bayes Optimal Classifier.
Approach

- Collect labeled email data.
- Extract features such as keywords and sender information.
- Train a Naïve Bayes classifier.
- Compare **Maximum Likelihood Estimation (MLE)** with the **Minimum Description Length (MDL)** principle.

Analysis

- Naïve Bayes computes spam probability.
- MLE selects parameters based only on training data.
- MDL chooses a simpler model to reduce overfitting.
- Larger datasets improve prediction accuracy.

Advantages

- Fast classification.
- Works well for text data.
- Simple implementation.

Limitations

- Assumes independent features.
- Performance decreases with correlated words.
- Requires sufficient training data.

Result

The spam detection system accurately classifies emails, and MDL provides better generalization than MLE in many cases.

3. Customer Segmentation Using EM Algorithm

Aim

To segment customers using the Expectation-Maximization (EM) algorithm.

Approach

- Collect customer purchase data.
- Initialize cluster parameters.
- Apply the EM algorithm.
- Identify customer groups based on purchasing behavior.

Analysis

- The **Expectation step** estimates cluster membership probabilities.
- The **Maximization step** updates cluster parameters.
- Different initial values may produce different clusters.
- EM handles missing or incomplete data effectively.

Advantages

- Finds hidden customer groups.
- Handles uncertainty.
- Works well with incomplete data.

Limitations

- Sensitive to initialization.
- May converge to local optima.
- Computationally expensive.

Result

The EM algorithm successfully groups customers into meaningful clusters for targeted marketing.

4. Medical Image Classification Using Candidate Elimination

Aim

To classify medical images as normal or abnormal using Candidate Elimination.

Approach

- Use labeled medical images.
- Define the hypothesis space.
- Apply the Candidate Elimination algorithm.
- Obtain specific and general hypotheses.

Analysis

- The algorithm updates hypotheses using positive and negative examples.
- A finite hypothesis space learns faster.
- Infinite hypothesis spaces require more training examples.
- More samples improve generalization.

Advantages

- Produces consistent hypotheses.
- Learns from examples.
- Easy to understand.

Limitations

- Sensitive to noisy data.
- Large hypothesis spaces increase complexity.
- Not suitable for continuous attributes without preprocessing.

Result
The Candidate Elimination algorithm effectively learns classification rules and generalizes to unseen medical images.

5. Investment Decision System Using Bayesian Inference
Aim
To predict investment outcomes using Bayesian inference, the Gibbs algorithm, and probability learning.
Approach
- Collect historical market data.
- Set prior probabilities based on past trends.
- Use Bayesian inference to update predictions.
- Apply the Gibbs algorithm to estimate complex posterior distributions.

Analysis
- Prior probabilities represent initial market beliefs.
- Posterior probabilities are updated as new data arrives.
- The Gibbs algorithm improves estimation when direct computation is difficult.
- The model adapts to changing market conditions.

Advantages
- Handles uncertainty effectively.
- Continuously updates predictions.
- Supports better investment decisions.

Limitations
- Computationally intensive.
- Requires good prior knowledge.
- Sensitive to poor-quality data.

Result
The Bayesian model provides reliable investment predictions by continuously updating beliefs with new market information, helping reduce decision-making risk.

6. Autonomous Vehicle Decision System
Aim
To design a probabilistic model for safe driving decisions using Bayesian Inference and Naïve Bayes.
Approach
- Collect data from sensors (camera, radar, LiDAR, GPS).
- Use **Bayesian Inference** to update the probability of road conditions.
- Apply **Naïve Bayes** to classify safe driving actions (Stop, Slow Down, Turn, Move).
- Select the action with the highest probability.

Analysis
- Bayesian inference updates predictions based on new sensor data.
- Naïve Bayes quickly predicts driving actions.
- Sensor noise and uncertainty are handled using probability values.
- The model adapts to changing weather and traffic conditions.

Advantages
- Fast decision making.
- Handles uncertain sensor data.
- Improves driving safety.

Limitations

- Assumes feature independence.
- Requires accurate sensor data.
- Performance decreases if sensors fail.

Result

The probabilistic model enables autonomous vehicles to make safe and reliable driving decisions under uncertain conditions.

7. Chronic Disease Risk Prediction

Aim

To predict patient risk levels for chronic diseases using Bayes' Theorem and probability learning.

Approach

- Collect patient history and lifestyle data.
- Calculate prior probabilities of diseases.
- Apply **Bayes' Theorem** to estimate disease risk.
- Use hypothesis space search to identify the best prediction model.

Analysis

- Prior probabilities influence predictions.
- Larger datasets improve prediction accuracy.
- Feature dependencies affect disease risk estimation.
- The model provides interpretable results for doctors.

Advantages

- Early disease detection.
- Supports preventive healthcare.
- Easy to interpret.

Limitations

- Requires quality medical data.
- Correlated features may reduce accuracy.
- Limited by incorrect prior probabilities.

Result

The Bayesian model effectively predicts chronic disease risk and supports preventive healthcare planning.

8. Fraud Detection in Banking

Aim

To detect fraudulent banking transactions using Bayesian Belief Networks (BBN) and the EM algorithm.

Approach

- Collect transaction history.
- Build a Bayesian Belief Network to model relationships between transaction features.
- Use the EM algorithm to estimate hidden fraud patterns.
- Predict whether a transaction is genuine or fraudulent.

Analysis

- BBN models dependencies among transaction attributes.
- EM handles incomplete and missing transaction data.
- Hidden variables improve fraud detection.
- Probability reasoning reduces false positives.

Advantages

- Detects hidden fraud patterns.
- Handles incomplete data.
- Improves detection accuracy.

Limitations

- Computationally expensive.
- Requires sufficient training data.
- Sensitive to initialization.

Result

The proposed system accurately identifies fraudulent transactions while minimizing false alarms.

9. Product Recommendation System

Aim

To recommend products using Naïve Bayes and Bayesian Learning.

Approach

- Collect user browsing and purchase history.
- Calculate the probability of user interest in products.
- Recommend products with the highest probability.
- Continuously update recommendations as user behavior changes.

Analysis

- Conditional probabilities determine product recommendations.
- Bayesian learning updates user preferences over time.
- The model handles uncertainty in customer interests.
- Suitable for large e-commerce platforms.

Advantages

- Personalized recommendations.
- Fast prediction.
- Easy to scale.

Limitations

- Assumes independent features.
- Requires sufficient user history.
- Cold-start problem for new users.

Result

The recommendation system provides personalized product suggestions and improves customer satisfaction.

10. Climate Change Prediction System

Aim

To predict climate changes using Bayesian Inference and the EM algorithm.

Approach

- Collect historical climate data (temperature, rainfall, humidity).
- Apply Bayesian inference to update predictions.
- Use the EM algorithm to estimate missing environmental data.
- Forecast future climate conditions.

Analysis

- Bayesian inference incorporates prior environmental knowledge.
- EM estimates missing values in climate datasets.
- The model handles noisy and incomplete data.
- Convergence depends on initialization.

Advantages

- Handles uncertain environmental data.
- Improves long-term forecasting.
- Adapts to new observations.

Limitations

- High computational cost.
- Forecast accuracy depends on data quality.
- Sensitive to incorrect prior assumptions.

Result

The probabilistic model effectively predicts climate changes by handling uncertainty and incomplete data, making it useful for environmental planning and decision-making.

11. Customer Churn Prediction System

Aim

To predict customer churn using Naïve Bayes, Bayesian Inference, and hypothesis space search.

Approach

- Collect customer usage and service feedback data.
- Calculate prior probabilities of customer churn.
- Train a Naïve Bayes classifier.
- Use Bayesian inference to update predictions with new customer information.

Analysis

- Prior probabilities improve churn prediction.
- Feature dependencies such as service quality and usage affect results.
- Uncertainty in customer behavior is handled using probability values.
- Larger datasets improve prediction accuracy.

Advantages

- Early identification of customers likely to leave.
- Improves customer retention strategies.
- Fast and easy to implement.

Limitations

- Assumes feature independence.
- Requires sufficient historical data.
- Accuracy depends on data quality.

Result

The Bayesian-based churn prediction system effectively identifies customers at risk of leaving, helping organizations improve customer retention.

12. Smart Healthcare Monitoring System

Aim

To monitor patients using wearable sensors with Bayesian Belief Networks and probability learning.

Approach

- Collect real-time vital signs such as heart rate, temperature, and blood pressure.
- Construct a Bayesian Belief Network.
- Detect abnormal health conditions using probability learning.
- Alert doctors when abnormal patterns are detected.

Analysis
- BBN models relationships among vital signs.
- Sensor noise is handled using probabilistic reasoning.
- Real-time updates improve prediction accuracy.
- Supports continuous patient monitoring.

Advantages
- Early detection of health issues.
- Handles uncertain sensor readings.
- Supports real-time healthcare.

Limitations
- Requires reliable wearable sensors.
- High computational requirements.
- Data privacy concerns.

Result
The smart healthcare system accurately detects abnormal health conditions and provides timely alerts for better patient care.

13. Logistics Route Optimization System

Aim
To optimize delivery routes using Bayesian Inference and probabilistic reasoning.

Approach
- Collect traffic, weather, and road condition data.
- Use Bayesian inference to predict the best route.
- Update route predictions dynamically using real-time information.
- Recommend the optimal delivery path.

Analysis
- Prior knowledge provides an initial route estimate.
- Real-time updates improve decision-making.
- The system handles uncertainty caused by traffic and weather.
- Computational complexity increases with larger road networks.

Advantages
- Reduces delivery time.
- Saves fuel costs.
- Adapts to changing road conditions.

Limitations
- Requires continuous real-time data.
- High processing requirements.
- Dependent on GPS and traffic information.

Result
The Bayesian route optimization system efficiently predicts the best delivery routes, improving logistics performance.

14. Cybersecurity Intrusion Detection System

Aim
To detect network intrusions using Naïve Bayes, EM Algorithm, and Bayesian Learning.

Approach
- Collect network traffic data.
- Train a Naïve Bayes classifier to classify normal and abnormal traffic.
- Use the EM algorithm to identify hidden attack patterns.
- Continuously update the model using Bayesian learning.

Analysis

- Naïve Bayes provides fast intrusion detection.
- EM identifies hidden patterns in incomplete data.
- Bayesian learning improves the model over time.
- The system reduces false alarms.

Advantages

- Fast detection of cyber attacks.
- Handles incomplete and uncertain data.
- Improves detection accuracy.

Limitations

- Sensitive to noisy network data.
- Requires frequent model updates.
- Computational cost increases with network size.

Result

The intrusion detection system effectively identifies network attacks while reducing false positives and improving cybersecurity.

15. Crop Yield Prediction System

Aim

To predict crop yield using Bayesian Inference and probability learning.

Approach

- Collect agricultural data such as rainfall, soil quality, and temperature.
- Apply Bayesian inference to estimate crop yield.
- Update predictions using seasonal and historical data.
- Recommend suitable farming strategies.

Analysis

- Prior agricultural knowledge improves predictions.
- Seasonal changes influence crop yield.
- Probability learning continuously updates the model.
- The system handles uncertainty in environmental conditions.

Advantages

- Supports farmers in planning cultivation.
- Improves agricultural productivity.
- Assists government policy planning.

Limitations

- Depends on accurate environmental data.
- Extreme weather conditions reduce prediction accuracy.
- Requires regular data updates.

Result

The Bayesian crop yield prediction model successfully estimates crop production under uncertain environmental conditions, supporting farmers and policymakers in making informed agricultural decisions.